

Residue at a Pole

Laurent's series, ~~expansion~~,

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$$

Let, $z=a$ be a pole of order m of a function $f(z)$ and C_1 circle of radius δ with centre at $z=a$, which does not contain any other singularities except at $z=a$.

Then, $f(z)$ is analytic within ^{annular} ~~analyse~~ region,

$$\delta < |z-a| < R$$

can be expanded within ^{annular} ~~analyse~~ region by Laurent's series.

$$f(z) = [a_0 + a_1(z-a) + a_2(z-a)^2 + \dots]$$

$$+ \left[\frac{b_1}{(z-a)} + \frac{b_2}{(z-a)^2} + \dots \right]$$

where, $a_n = \frac{1}{2\pi i} \int_{C_2} \frac{f(z)}{(z-a)^{n+1}} dz$, $|z-a|=R$ being circle C_2 .

$$b_n = \frac{1}{2\pi i} \int_{C_1} \frac{f(z)}{(z-a)^{-n+1}} dz, |z-a|=\delta \text{ being circle } C_1.$$

then,

$$b_1 = \frac{1}{2\pi i} \int_{C_1} f(z) dz$$

The coefficient b_1 is called Residue of $f(z)$ at $z=a$.

$$\text{Res}(z=a) = b_1 = \frac{1}{2\pi i} \int_{C_1} f(z) dz.$$

Methods to Find Residue

1. If $z=a$ is a simple pole,

$$(a) \text{Res}(z=a) = \lim_{z \rightarrow a} \left\{ (z-a) f(z) \right\}.$$

$$(b) \text{ If } f(z) = \frac{g(z)}{h(z)}$$

where $h(a)=0$ & $g(a) \neq 0$,

$$\text{Res}(z=a) = \frac{g(a)}{h'(a)}.$$

2. If $z=a$ is pole of order n ,

$$\text{Res}(z=a) = \frac{1}{(n-1)!} \left[\frac{d^{n-1}}{dz^{n-1}} \left\{ (z-a)^n f(z) \right\} \right]_{z=a}$$

3. If $z=a$ is a pole,

$$f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \sum_{n=1}^{\infty} b_n (z-a)^{-n}$$

$$\Rightarrow f(z) = \sum_{n=0}^{\infty} a_n (z-a)^n + \left[\frac{b_1}{(z-a)} + \frac{b_2}{(z-a)^2} + \dots + \frac{b_m}{(z-a)^m} \right]$$

$$\text{Res}(z=a) = \text{Coeff. of } \frac{1}{z-a} = b_1$$

4. Residue at $z = \infty$,

$$\text{Res}(z = \infty) = \lim_{z \rightarrow \infty} \int -z f(z)$$

Residue Theorem

If $f(z)$ is analytic inside a simple closed curve C and on C , except at a finite number of poles (singularities) $z_1, z_2, \dots, z_n$ inside C then,

$$\int_C f(z) dz = 2\pi i (\text{Sum of Residue at the point poles lie inside } C).$$

Q.1) Determine the poles of the function,
 $f(z) = \frac{z^2}{(z-1)^2(z+2)}$ and the residue at each pole. Hence evaluate $\int_C \frac{z^2}{(z-1)^2(z+2)} dz$ where C is $|z| = 5/2$.

Solⁿ:

For poles,

$$(z-1)^2(z+2) = 0$$
$$\Rightarrow z = 1, -2$$

Here, $z=1$ is pole of order 2
& $z=-2$ is pole of order 1 (i.e. simple pole).

$$\text{Res}(z=1) = \frac{1}{(2-1)!} \left[\frac{d}{dz} \left\{ \frac{z^2}{(z-1)^2(z+2)} \right\} \right]_{z=1}$$

$$\Rightarrow \text{Res}(z=1) = \frac{1}{1!} \left[\frac{d}{dz} \left(\frac{z^2}{z+2} \right) \right]_{z=1}$$

$$= \left[\frac{(z+2) \cdot 2z - z^2}{(z+2)^2} \right]_{z=1}$$

$$= \frac{(1+2) \cdot 2 \times 1 - 1^2}{(1+2)^2}$$

$$= \frac{6-1}{9}$$

$$\Rightarrow \text{Res}(z=1) = \frac{5}{9}$$

Now,

$$f(z) = \frac{z^2}{(z-1)^2(z+2)} = \frac{g(z)}{h(z)}$$

$$g(z) = z^2 \Rightarrow g(-2) = 4 \neq 0$$

$$h(z) = (z-1)^2(z+2) \Rightarrow h(-2) = 0$$

$$\text{Res}(z=-2) = \frac{g(-2)}{h'(-2)}$$

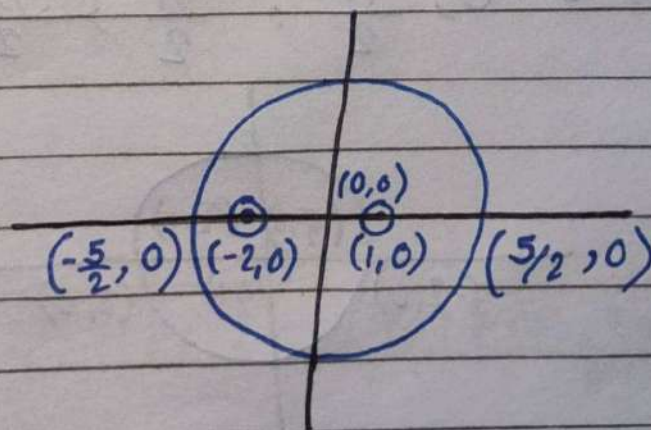
$$h'(z) = (z-1)^2 + 2(z-1)(z+2)$$

$$\Rightarrow h'(-2) = (-2-1)^2 + 2(-2-1)(-2+2) = 9$$

$$\therefore \text{Res}(z=-2) = \frac{g(-2)}{h'(-2)} = \frac{4}{9}$$

$$\int_C \frac{z^2}{(z-1)^2(z+2)} dz = 2\pi i (\text{Sum of residues at the poles lies inside } C)$$

$$|z| = 5/2$$



$\therefore z=1, -2$ lies inside C .

$$\therefore \int_C \frac{z^2}{(z-1)^2(z+2)} dz = 2\pi i [\text{Res}(z=1) + \text{Res}(z=-2)]$$

$$= 2\pi i \left[\frac{5}{9} + \frac{4}{9} \right]$$

$$= 2\pi i (1)$$

$$= 2\pi i.$$

Q.2) Find the sum of residues of the function $f(z) = \frac{\sin z}{z \cos z}$ at its poles inside the circle $|z|=2$.

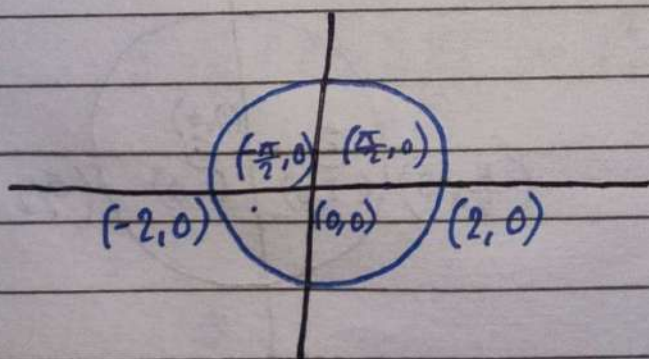
Solⁿ:

For poles,

$$z \cos z = 0$$

$$\Rightarrow z=0, \cos z=0 \Rightarrow z=(2n \pm 1)\pi/2$$

$$\Rightarrow z=0, \frac{\pi}{2}, -\frac{\pi}{2}, \frac{3\pi}{2}, \dots$$



Here, $z = 0, \frac{\pi}{2}, -\frac{\pi}{2}$ lies inside C .

$$f(z) = \frac{\sin z}{z \cos z} = \frac{g(z)}{h(z)}$$

$$g(z) = \sin z$$

$$h(z) = z \cos z \Rightarrow h'(z) = -z \sin z + \cos z$$

For $z = 0$,

$$g(0) = \sin 0 = 0$$

$$h(0) = 0$$

$$\therefore g(0) = 0 \text{ \& } h(0) = 0$$

$$\therefore \text{Res}(z=0) = \lim_{z \rightarrow 0} \left[z \cdot \frac{\sin z}{z \cos z} \right]$$

$$= \lim_{z \rightarrow 0} [\tan z]$$

$$= 0$$

For $z = \pi/2$,

$$g(\pi/2) = \sin \pi/2 = 1 \neq 0$$

$$h(\pi/2) = 0$$

$$\therefore \text{Res}(z = \pi/2) = \frac{g(\pi/2)}{h'(\pi/2)} = \frac{1}{[-\pi/2 \sin(\pi/2) + \cos(\pi/2)]} = \frac{-2}{\pi}$$

For $z = -\pi/2$,

$$\begin{aligned}\operatorname{Res}(z = -\pi/2) &= \frac{g(\pi/2)}{h'(-\pi/2)} = \frac{-1}{[\pi/2 \sin(-\pi/2) + \cos(-\pi/2)]} \\&= \frac{-1}{-\pi/2} \\&= \frac{2}{\pi}.\end{aligned}$$

Hence, required sum is,

$$\operatorname{Res}(z=0) + \operatorname{Res}(z=\pi/2) + \operatorname{Res}(z=-\pi/2)$$

$$= 0 - \frac{2}{\pi} + \frac{2}{\pi}$$

$$= 0.$$

Q.3) Determine the poles of the following function and residue at each pole $f(z) = \frac{(z-1)}{(z+1)^2(z-2)}$ and

hence evaluate $\oint_C f(z) dz$, where C is a circle

$$|z-i| = 2.$$

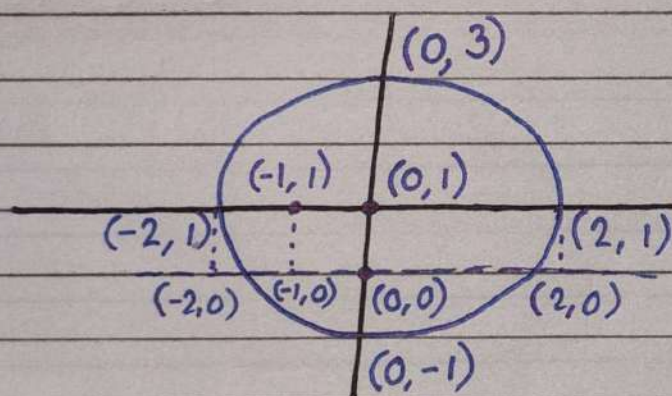
Solⁿ:

For poles,

$$(z+1)^2(z-2) = 0$$

$\Rightarrow z = -1$ is pole of order 2.

& $z = 2$ is simple pole.



$$|z-i| = 2$$

Centre = $(0, 1)$
Radius = 2.

$\therefore z = -1$ lies inside C .

$$\therefore \int_C f(z) dz = 2\pi i (\text{Res}(z = -1))$$

$$\text{Res}(z = -1) = \frac{1}{(2-1)!} \left[\frac{d}{dz} \left\{ \frac{(z-1)}{(z+1)^2(z-2)} \right\} \right]_{z=-1}$$

$$= \frac{1}{1!} \left[\frac{d}{dz} \frac{(z-1)}{(z-2)} \right]_{z=-1}$$

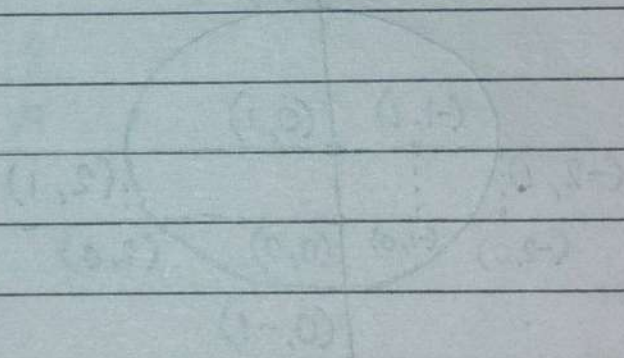
$$\Rightarrow \text{Res}(z=-1) = \left[\frac{(z-2) - (z-1)}{(z-2)^2} \right]_{z=-1}$$

$$= \frac{(-1-2) - (-1-1)}{(-1-2)^2}$$

$$= \frac{-3+2}{9}$$

$$= -\frac{1}{9}$$

$$\therefore \int_c f(z) dz = -\frac{2\pi i}{9}$$



Q.1) Evaluate $\int_C \frac{z^2 - 2z}{(z+1)^2(z^2+4)} dz$, where C is the circle $|z| = 10$, by Residue Theorem.

Solⁿ:

For poles,

$$(z+1)^2(z^2+4) = 0$$
$$\Rightarrow z = -1, \pm 2i$$

$z = -1, \pm 2i$ lies inside $|z| = 10$.

$\therefore z = -1$ is pole of order 2.

$$\therefore \text{Res}(z=-1) = \frac{1}{(2-1)!} \left[\frac{d^{2-1}}{dz^{2-1}} \left[\frac{(z+1)^2}{(z+1)^2(z^2+4)} \cdot \frac{z^2-2z}{(z+1)^2(z^2+4)} \right] \right]_{z=-1}$$

$$= \frac{1}{1!} \left[\frac{d}{dz} \left(\frac{z^2-2z}{z^2+4} \right) \right]_{z=-1}$$

$$= \left[\frac{(2z-2)(z^2+4) - (z^2-2z)(2z)}{(z^2+4)^2} \right]_{z=-1}$$

$$= \frac{(2(-1)-2)((-1)^2+4) - ((-1)^2-2(-1))(2(-1))}{((-1)^2+4)^2}$$

$$= \frac{(-4 \times 5) - (3 \times -2)}{25}$$

$$= \frac{-20+6}{25} = \frac{-14}{25}$$

$\therefore z = 2i$ is a simple pole.

$$\text{Res}(z = 2i) = \frac{g(2i)}{h'(2i)}$$

$$g(z) = z^2 - 2z \Rightarrow g(2i) = -4 - 4i \neq 0$$

$$h(z) = (z+1)^2(z^2+4) \Rightarrow h(2i) = 0$$

~~$$\text{Res}(z = 2i) \quad h'(z) = 2(z+1) \cdot z + (z+1)^2 \cdot 2z$$~~

$$h'(z) = (z+1)^2 \cdot 2z + 2(z+1)(z^2+4)$$

$$\text{Res}(z = 2i) = \frac{g(2i)}{h'(2i)} = \frac{-4-4i}{2(2i)(2i+1)^2 + 0}$$

$$= \frac{-4-4i}{4i(2i+1)^2}$$

$$= \frac{-(1+i)}{i(4+4i+1)}$$

$$= \frac{-(1+i)}{i(4i-3)}$$

$$= \frac{-(1+i)}{-3i-4}$$

$$= \frac{-(1+i)}{-(4+3i)} \times \frac{(4-3i)}{(4-3i)}$$

$$= \frac{(1+i)(4-3i)}{(4^2 - (3i)^2)}$$

$$= \frac{4 - 3i^2 - 3i + 4i}{16 + 9}$$

$$= \frac{4 + 3 - 3i + 4i}{25}$$

$$= \frac{7 + i}{25}$$

$\therefore z = -2i$ is a simple pole.

$$\text{Res}(z = -2i) = \frac{g(-2i)}{h'(-2i)}$$

$$g(z) = z^2 - 2z \Rightarrow g(-2i) = -4 + 4i \neq 0$$

&

$$h(z) = (z+1)^2(z^2+4) \Rightarrow h(-2i) = 0$$

Then,

$$h'(z) = (z+1)^2 \cdot 2z + 2(z+1)(z^2+4)$$

$$\Rightarrow h'(-2i) = 2(-2i)((-2i)+1)^2 + 0$$

$$= -4i(-2i+1)^2$$

$$= -4i(-4 - 4i + 1)$$

$$= -4i(-4i - 3)$$

$$\Rightarrow h(-2i) = 4i(4i+3)$$

$$\therefore \text{Res}(z = -2i) = \frac{Q(-2i)}{h'(-2i)} = \frac{-4+4i}{4i(4i+3)} = \frac{i-1}{i(4i+3)}$$

$$\Rightarrow \text{Res}(z = -2i) = \frac{i-1}{(-4+3i)}$$

$$= \frac{i-1}{3i-4} \times \frac{3i+4}{3i+4}$$

$$= \frac{3i^2+4i-3i-4}{(3i)^2-(4)^2}$$

$$= \frac{-3+i-4}{-9-16}$$

$$= \frac{i-7}{-25}$$

$$= \frac{7-i}{25}$$

$$\therefore \int_C \frac{z^2-2z}{(z+1)^2(z^2+4)} dz = 2\pi i \left(\frac{-14}{25} + \frac{7+i}{25} + \frac{7-i}{25} \right)$$

$$= \frac{2\pi i}{25} (\cancel{-14} + \cancel{7} + \cancel{7} + i - i)$$

$$= 0.$$

Q.2) Find the residue of $f(z) = \frac{z^3}{(z-1)^4(z-2)(z-3)}$ at the pole $z=1$.

Solⁿ:

Here, $z=1$ is a pole of order 4.

$$\begin{aligned}\text{Res}(z=1) &= \frac{1}{(4-1)!} \left[\frac{d^{4-1}}{dz^{4-1}} \left\{ \frac{z^3}{(z-1)^4(z-2)(z-3)} \right\} \right]_{z=1} \\ &= \frac{1}{3!} \left[\frac{d^3}{dz^3} \left(\frac{z^3}{(z-2)(z-3)} \right) \right]_{z=1}\end{aligned}$$

~~Simplifying~~, Simplifying,

$$\frac{z^3}{(z-2)(z-3)} = \frac{z^3}{z^2 - 5z + 6}$$

$$\frac{z^3}{z^2 - 5z + 6} = (z+5)$$

$$\begin{array}{r} z^3 - 5z^2 + 6z \\ \hline \end{array}$$

$$\begin{array}{r} 5z^2 - 6z \\ \hline \end{array}$$

$$\begin{array}{r} 5z^2 - 25z + 30 \\ \hline \end{array}$$

$$\underline{19z - 30}$$

$$\frac{z^3}{(z-2)(z-3)} = (z+5) + \frac{19z - 30}{(z-2)(z-3)}$$

$$\frac{19z-30}{(z-2)(z-3)} = \frac{A}{z-2} + \frac{B}{z-3}$$

$$\Rightarrow 19z - 30 = A(z-3) + B(z-2)$$

when $z=3$,

$$B = 19 \times 3 - 30 = 27$$

when $z=2$,

$$-A = 19 \times 2 - 30 = 8$$

$$\Rightarrow A = -8$$

~~Thus~~ So, $\frac{z^3}{(z-2)(z-3)} = z + 5 - \frac{8}{z-2} + \frac{27}{z-3}$

Now, $\frac{d}{dz} \left[\frac{z^3}{(z-2)(z-3)} \right] = 1 + \frac{8}{(z-2)^2} - \frac{27}{(z-3)^2}$

$$\frac{d^2}{dz^2} \left[\frac{z^3}{(z-2)(z-3)} \right] = \frac{-16}{(z-2)^3} + \frac{54}{(z-3)^3}$$

$$\frac{d^3}{dz^3} \left[\frac{z^3}{(z-2)(z-3)} \right] = \frac{48}{(z-2)^4} - \frac{162}{(z-3)^4}$$

So, $\text{Res}(z=1) = \frac{1}{3!} \left[\frac{d^3}{dz^3} \left(\frac{z^3}{(z-2)(z-3)} \right) \right]_{z=1}$

$$= \frac{1}{6} \left[\frac{48}{(z-2)^4} - \frac{162}{(z-3)^4} \right]_{z=1}$$

$$= \frac{1}{6} \left[\frac{48}{(1-2)^4} - \frac{162}{(1-3)^4} \right]$$

$$= \frac{1}{6} \left(48 - \frac{162}{8} \right)$$

$$= \frac{1}{6} \left(\frac{384 - 81}{8} \right)$$

$$= \frac{1}{6} \left(\frac{303}{8} \right)$$

$$= \frac{101}{16}$$

Hence, the residue of $f(z)$ at the pole ($z=1$) is $\frac{101}{16}$.